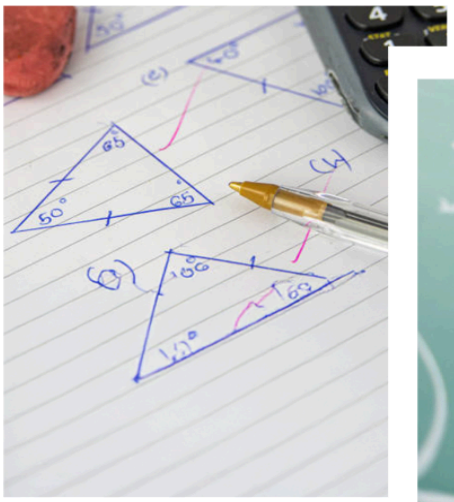
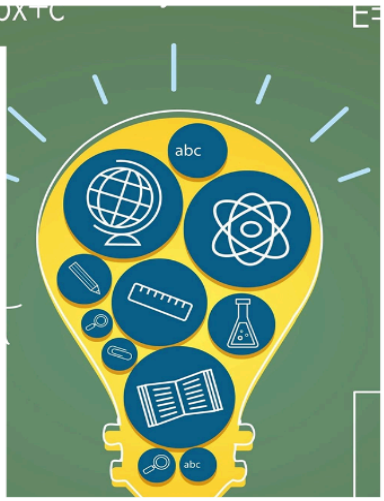
testbook PASS PRO MAX

Quantitative Aptitude

Formula – Guide

Interest





Interest

Simple Interest

- Simple Interest (S.I.) = $\frac{Principal \times Rate \times Time}{100}$ = $\frac{P \times R \times T}{100}$
- Solving for Principal : P = $\frac{SI \times 100}{R \times T}$
- Solving for Rate : R = $\frac{SI \times 100}{P \times T}$
- Solving for Time : T = $\frac{SI \times 100}{P \times R}$
- Total Amount = Principal + Simple Interest

  A = P + SI

Different Time Periods in Simple Interest

- Annual Interest: When the interest is calculated on a yearly basis.
- Semi-Annual and Quarterly Interest:

  T (for semi-annually) = $\frac{1}{2}$

  T (for quarterly) = $\frac{1}{4}$

Distinct rate of interest for the distinct time period

- Rate for 1st T_1 year = R_1%
- Rate for 2nd T_2 year = R_2%
- Rate for 3rd T_3 year = R_3%

  $SI = SI_1 + SI_2 + SI_3 + ...$

  $SI = P \times (R_1t_1 + R_2t_2 + R_3t_3 + ...) / 100$

Compound Interest

- CI = P × $\{(1 + \frac{R}{100n})^{nT} - 1\}$
- CI = A - P
- A = P × $(1 + \frac{R}{100n})^{nT}$

Types of Compounding:

	Rate	Time
Annually	R%	T
Half-yearly	R/2 %	2 × T
Quarterly	R/4 %	4 × T
Monthly	R/12 %	12 × T

The formula for amount in CI can be as follows:

- **Annually :** **A = P** $(1 + \frac{R}{100})^{T}$
- **Half-Yearly :** **A = P** $(1 + \frac{R/2}{100})^{2T}$
- **Quarterly :** **A = P** $(1 + \frac{R/4}{100})^{4T}$
- **Monthly :** **A = P** $(1 + \frac{R/12}{100})^{12\,T}$

Effective Annual Rate (EAR)

EAR = $(1 + \frac{R}{n})^{n} - 1$

Relation between CI & SI

For 2 years: CI - SI = P $(\frac{R}{100})^2$

Rate	CI	CI - SI
5%	10.25%	0.25%
10%	21%	1%
15%	32.25%	2.25%
20%	44%	4%
25%	56.25%	6.25%
30%	69%	9%

For 3 years: CI - SI = P $(\frac{R}{100})^2 \times (3 + \frac{R}{100})$

Rate	CI	CI - SI
5%	15.76%	0.76%
10%	33.1%	3.1%
15%	52.08%	7.08%
20%	72.8%	12.8%
25%	95.3%	20.3%
30%	119.7%	29.7%

Installment

- A = I × (n + r% × $\frac{n((n-1)}{2}$)
- A = I (1 + (1+r%) + (1+2r%) + (1+3r%) +.....

Where, I = first installment

I (1+r%) = second installment

I (1+2r%) = third installment

I (1+3r%) = fourth installment

Present Value Formula for Installments

$$P = \sum_{t=1}^{n} \left(\frac{A}{(1+r)^t}\right)$$

OR

$$P = \frac{A}{(1+r)^1} + \frac{A}{(1+r)^2} + \frac{A}{(1+r)^3} + \ldots\ldots$$

Future Value of Annuity (FVA)

$$FV = C \times \frac{(1+R)^T - 1}{R}$$